

Scratch

Firm type (`is_biotech`, `is_pharma`)

I initially planned to construct binary firm-type indicators using external biotech/pharma classifications. However, these categories are not mutually exclusive or consistently defined: many firms operate across both biotechnology and traditional pharmaceutical development. I therefore avoided imposing a potentially subjective firm-type label and used annual revenue as a continuous, consistently observable measure of firm size.

SIC codes as a biotech classifier

Where AI Was Wrong

Using SIC codes to classify biotechnology versus pharmaceutical firms.

The AI initially suggested that SEC SIC codes could be used to create separate `is_biotech` and `is_pharma` indicators. When I applied this approach, the classifications did not reflect the actual firm population. SIC 2834 included both small biotechnology companies and major pharmaceutical firms, while broader codes introduced irrelevant companies such as Coca-Cola. The actual output therefore showed that SIC codes could identify a broad life-sciences sample but could not reliably distinguish biotechnology from pharmaceutical firms. I revised the method by using SIC codes only as an industry filter and classifying firm size using annual revenue.

Hand-built biotech/pharma stock lists

I initially considered constructing `is_biotech` and `is_pharma` by manually identifying publicly traded biotech and pharmaceutical companies from stock-market/company lists. This would have provided a more intuitive firm-type classification than SIC codes, but it would require subjective decisions about which companies qualify as biotech versus pharmaceutical and would be difficult to apply consistently across the full 2016–2026 period.

Where AI Was Wrong

Interpreting missing firm data as random.

The AI initially treated observations without SEC revenue data as ordinary missing values that could simply be removed. However, missingness was systematically related to company characteristics: private and smaller firms were less likely to appear in SEC data and may also have been more involved in earlier-stage trials. I identified this as non-random sample selection rather than a routine data-cleaning issue.

Assuming firm-size categories covered every company.

The AI initially suggested assigning all companies to either small biotechnology or large pharmaceutical categories. The published revenue thresholds left a substantial intermediate group that did not satisfy either definition. Forcing those firms into a binary classification would have introduced arbitrary labels. I therefore retained an **OTHER** category and excluded it only when presenting the focused comparison between the two endpoint groups.

Private and non-U.S.-listed firms

Excluded due to data availability. I considered expanding the sample to private and non-U.S.-listed firms using alternative sources such as Wikidata and NLP-based inference from sponsor descriptions. I ultimately rejected these approaches because I wanted firm size to be based on a single, standardized, publicly auditable financial source rather than inferred values with uncertain measurement error. This creates an important coverage limitation: private and non-U.S.-listed firms are likely to be disproportionately represented among smaller and earlier-stage companies, meaning their exclusion may be systematically related to both the firm-size classification and the clinical-development outcome.

Revenue deflation and R&D-spending criterion

I considered both adjusting revenue for inflation and incorporating R&D spending into the definition of small biotechnology firms. Both were feasible extensions, but were excluded to maintain scope and because of time constraints. These are therefore among the first methodological extensions I would prioritize in future analysis.

Global revenue rankings / top-20 pharmaceutical companies

Rejected. I considered defining large pharmaceutical firms using a global revenue ranking or a hand-assembled list of the largest pharmaceutical companies. I instead used a published revenue threshold so that firm classification would be based on a transparent and reproducible rule, and to preserve meaningful variation in firm size within the analysis.

Employee headcount

I rejected this as a measure of firm size as, although employee headcount may conceptually capture organizational scale, it can change substantially. More importantly, consistently measured point-in-time headcount was not available across the full set of firms and years required for the time-varying design. Annual revenue therefore provided a more consistently available measure for this analysis.

Risk defined by drug novelty

I considered defining development risk using drug novelty or first-in-class status rather than clinical-trial phase. This would provide a more direct measure of development risk, but the available registry data were insufficient to construct a reliable novelty measure. I therefore use clinical-trial phase as a measure of development stage, rather than treating phase as a direct measure of risk.

Therapeutic indication

Therapeutic indication is potentially relevant because the mix of indications could influence the relationship between firm size and clinical-trial phase. However, I did not have sufficient time nor enough knowledge to analyze and normalize the AACT conditions data within this study. I therefore excluded indication from the current analysis and treat it as an important area for future work.

Formal testing

Due to time constraints, I did not conduct formal hypothesis testing, regression analysis, or other causal/statistical approaches to test the relationship between firm size and trial phase. Instead, I focused on a descriptive comparison of phase distributions.